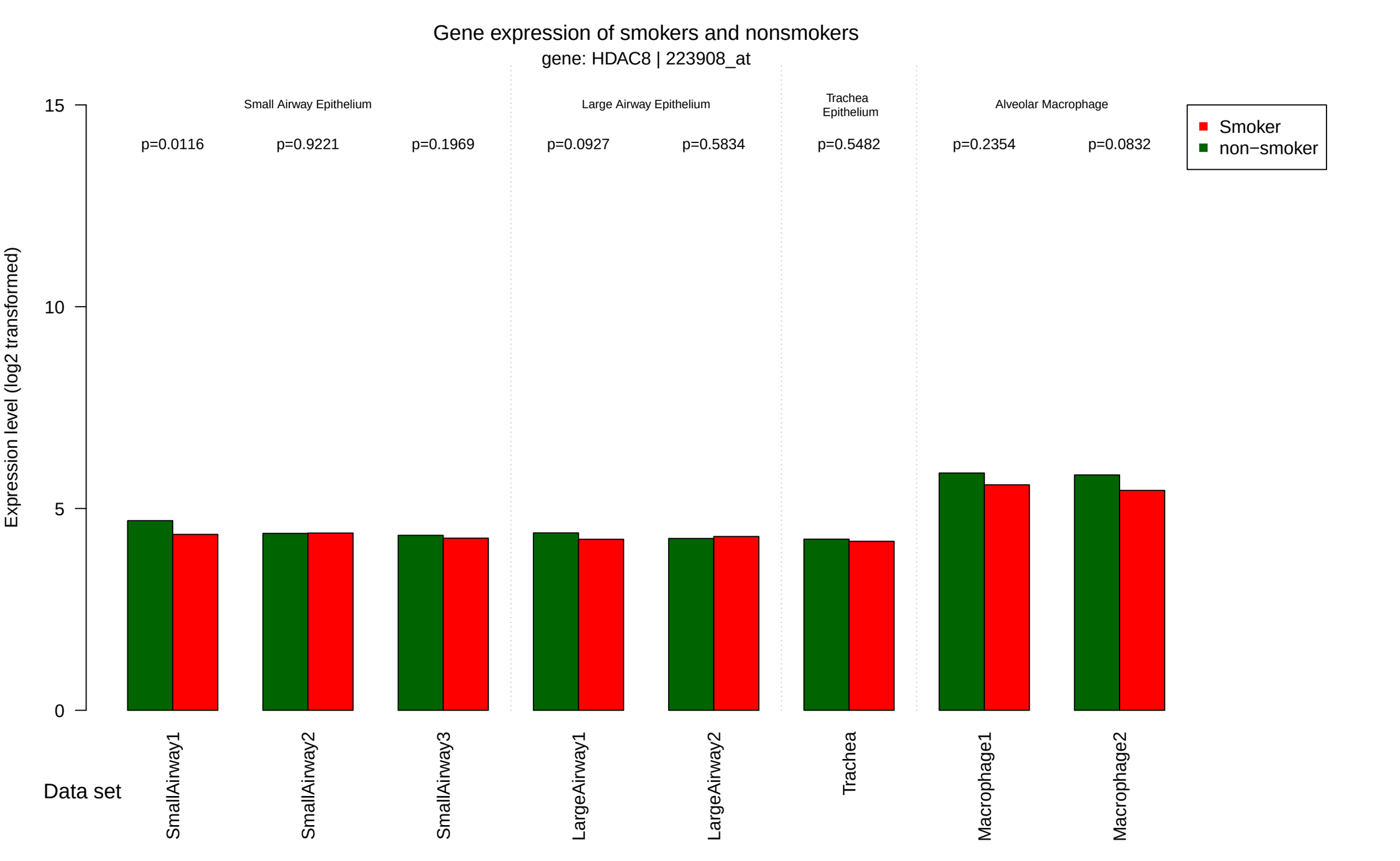
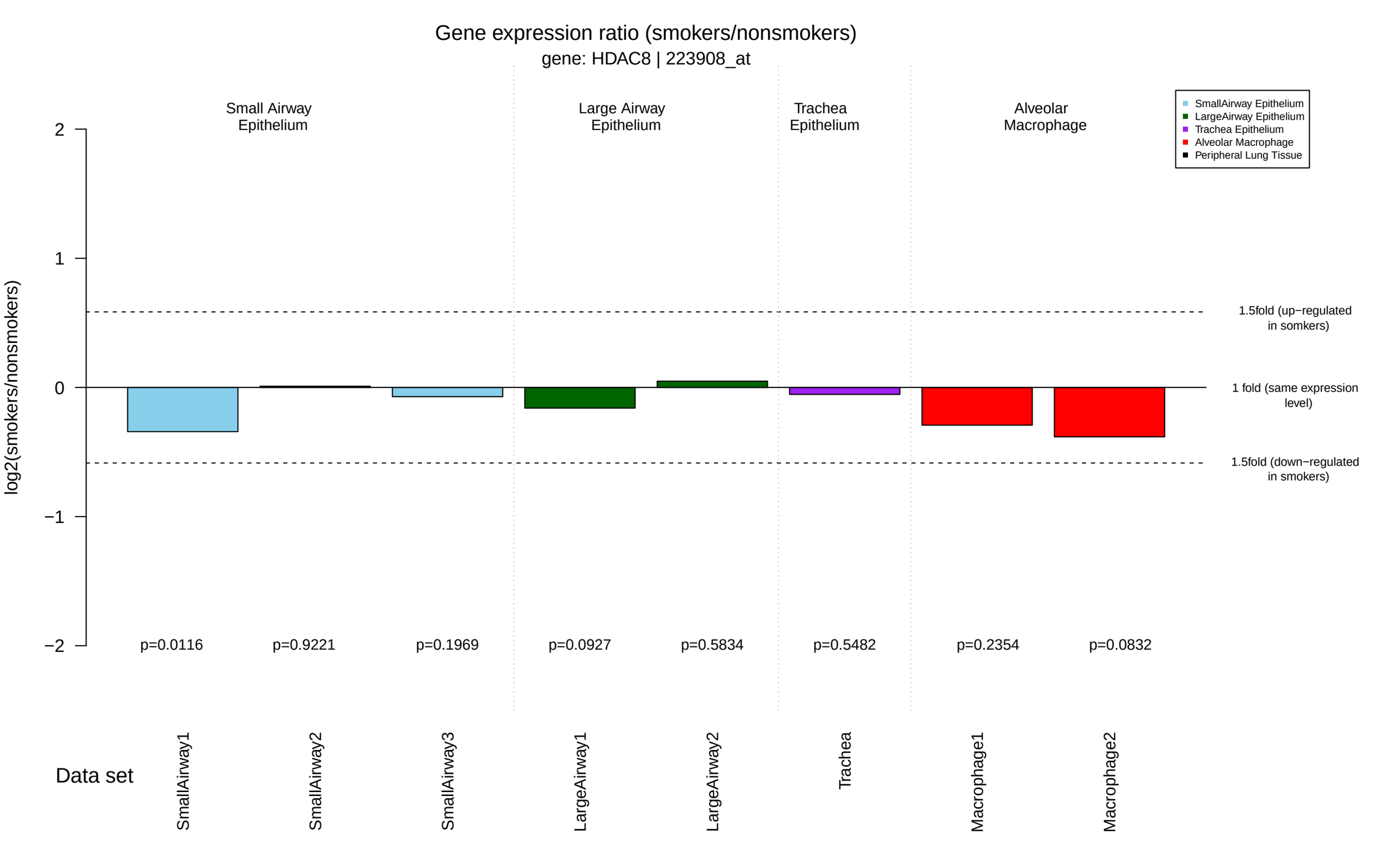
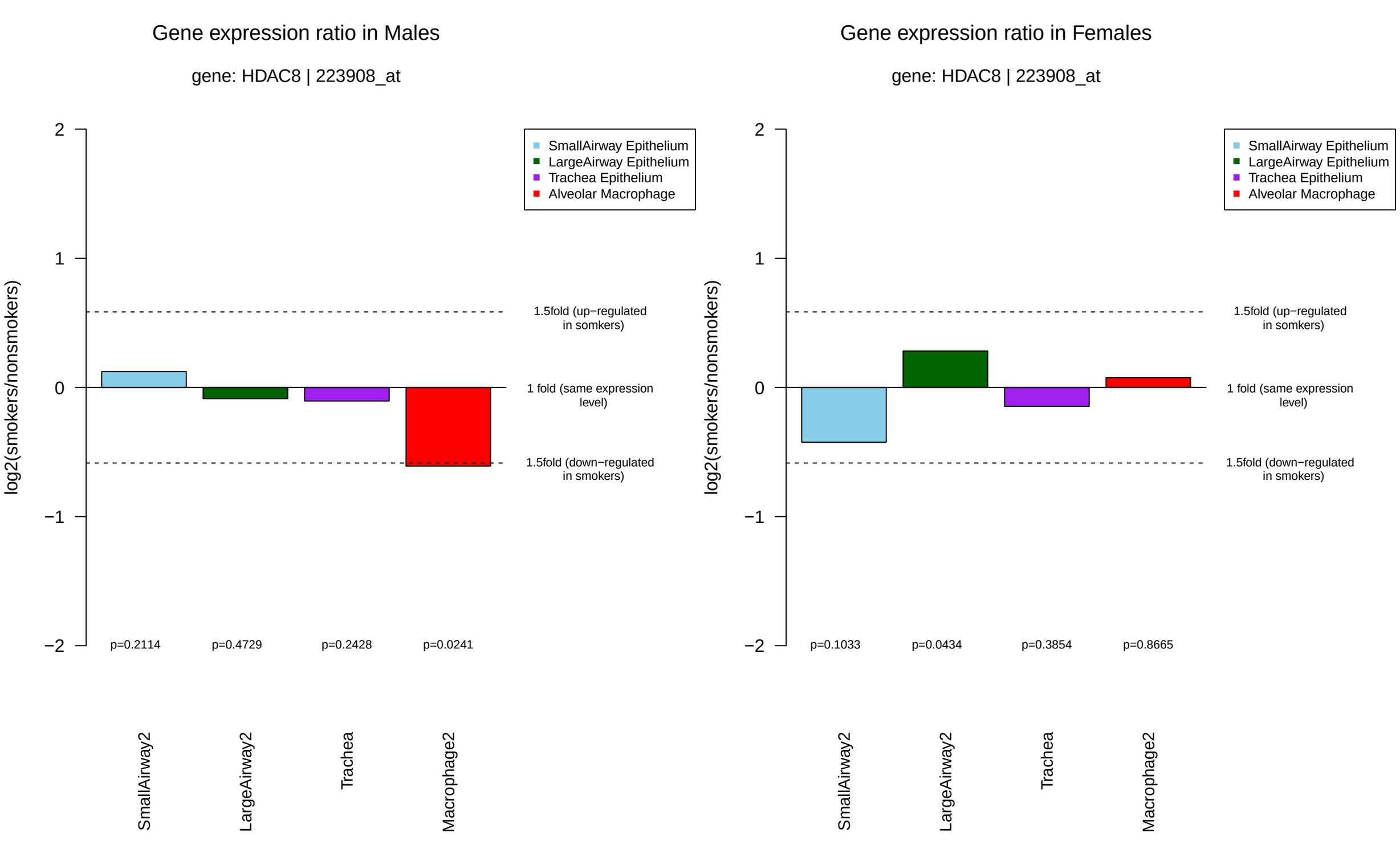
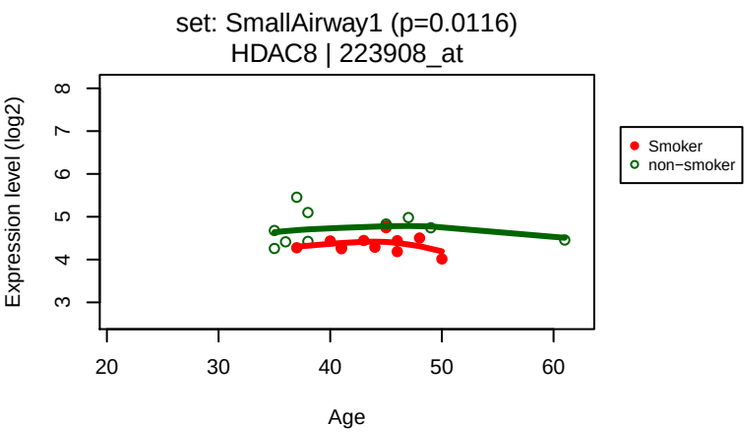
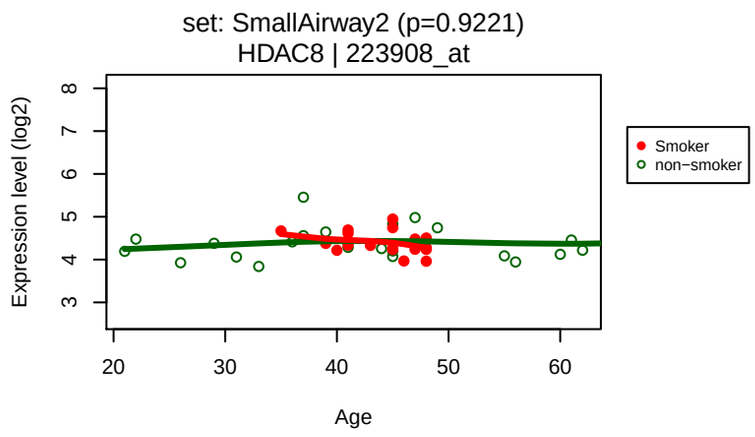
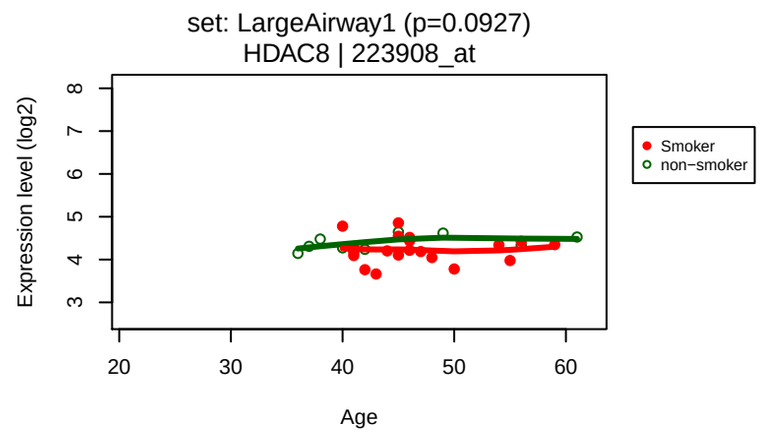

Gene expression of smokers and nonsmokers
gene: HDAC8 | 223908_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0116
p=0.9221
p=0.1969
p=0.0927
p=0.5834
p=0.5482
p=0.2354
p=0.0832
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

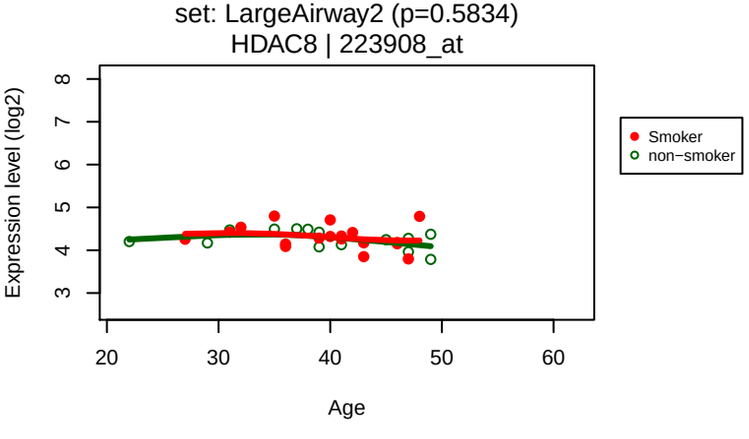

Gene expression ratio (smokers/nonsmokers)
gene: HDAC8 | 223908_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0116
p=0.9221
p=0.1969
p=0.0927
p=0.5834
p=0.5482
p=0.2354
p=0.0832
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

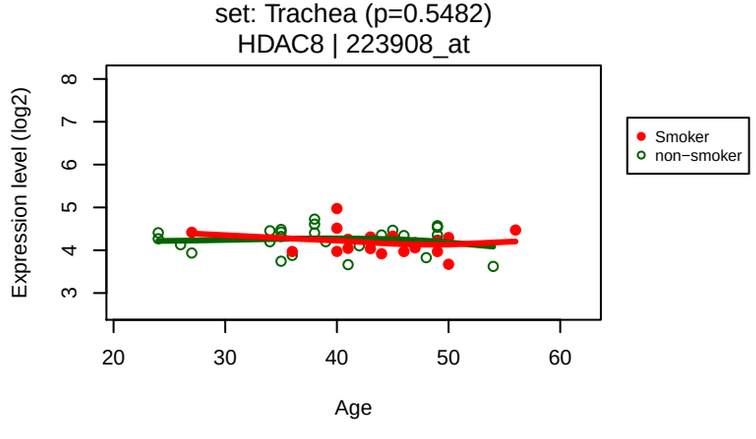

Gene expression ratio in Males
gene: HDAC8 | 223908_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2114
p=0.4729
p=0.2428
p=0.0241
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HDAC8 | 223908_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1033
p=0.0434
p=0.3854
p=0.8665
SmallAirway2
LargeAirway2
Trachea
Macrophage2

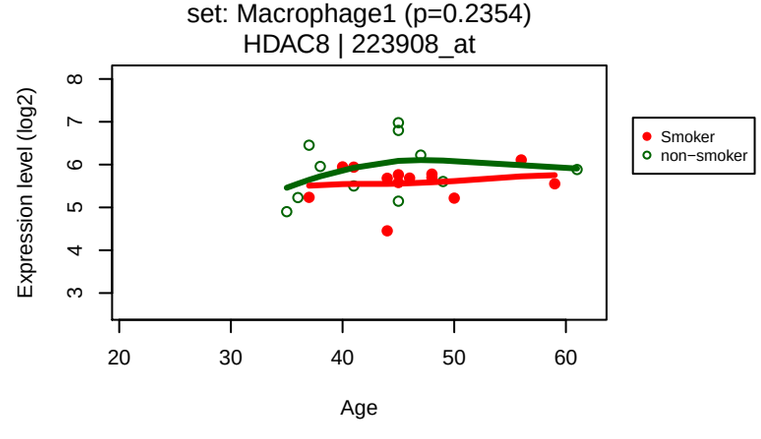

set: SmallAirway1 (p=0.0116)
HDAC8 | 223908_at
Expression level (log2)
Age
Smoker
non−smoker

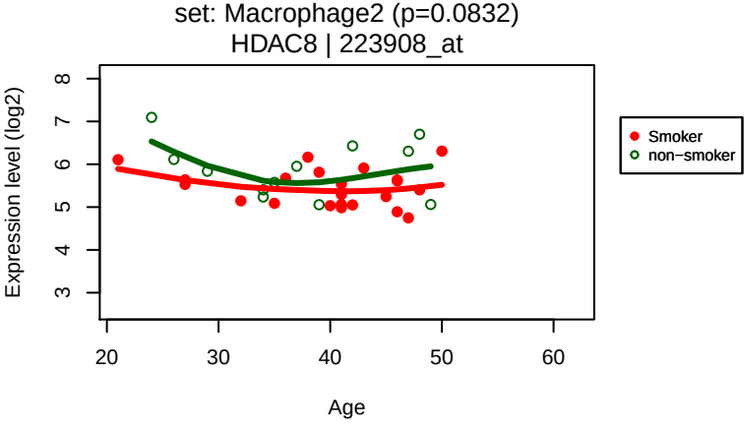

set: SmallAirway2 (p=0.9221)
HDAC8 | 223908_at
Expression level (log2)
Age
Smoker
non−smoker

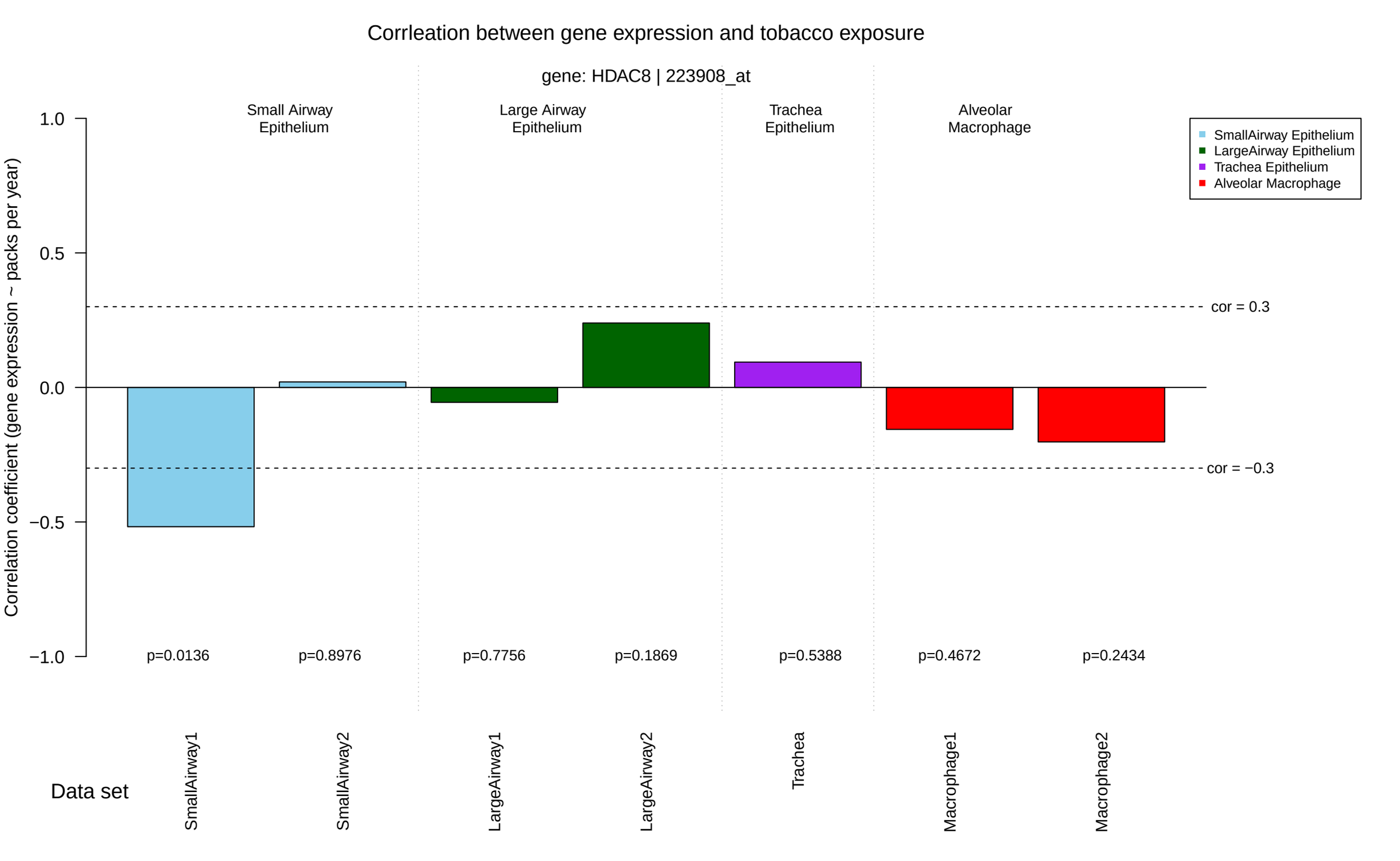

set: LargeAirway1 (p=0.0927)
HDAC8 | 223908_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5834)
HDAC8 | 223908_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5482)
HDAC8 | 223908_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2354)
HDAC8 | 223908_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0832)
HDAC8 | 223908_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HDAC8 | 223908_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0136
p=0.8976
p=0.7756
p=0.1869
p=0.5388
p=0.4672
p=0.2434
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2